Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: B-6 Dairy

Physical address of dairy:

22014 Turner AVE	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X200-X003-XXXX

B. OPERATORS

Borba, Mike

Operator name: Borba, Mike

Telephone no.: (209) 656-9942 (Landline) (209) 652-4079 (Cellular)

22014 Turner AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Borba, Connie

Legal owner name: Borba, Connie

Telephone no.: (209) 634-4930 (Landline)

5297 Kelley RD	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

Borba, Emanuel

Legal owner name: Borba, Emanuel

Telephone no.: (209) 634-4930 (Landline)

5297 Kelley RD	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

Borba, Evelyn

Legal owner name:	Borba, Evelyn	Telephone no.:	(209) 634-4930	
			Landline	Cellular
5297 Kelley RD	Hilmar		CA	95324
Mailing Address Number and Street	City		State	Zip Code

Borba, Johnny

Legal owner name:	Borba, Johnny	Telephone no.:	(209) 634-4930	
			Landline	Cellular
5297 Kelley RD	Hilmar		CA	95324
Mailing Address Number and Street	City		State	Zip Code

Borba, Maryana

Legal owner name:	Borba, Maryana	Telephone no.:	(209) 634-4930	
			Landline	Cellular
5297 Kelley RD	Hilmar		CA	95324
Mailing Address Number and Street	City		State	Zip Code

Borba, Mike

Legal owner name:	Borba, Mike	Telephone no.:	(209) 656-9942	(209) 652-4079
			Landline	Cellular
22014 Turner AVE	Hilmar		CA	95324
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1,652	212	140	115	0	0
Maximum number	1,652	212	140	115	0	0
Average number	1,652	212	140	115	0	0
Avg live weight (lbs)	1,500	1,550	1,100	800		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 48,585.05 *tons per reporting period*

Total nitrogen from manure: 622,007.28 *lbs per reporting period*

After ammonia losses (30% loss applied): 435,405.10 *lbs per reporting period*

Total phosphorus from manure: 103,931.57 *lbs per reporting period*

Total potassium from manure: 318,543.97 *lbs per reporting period*

Total salt from manure: 826,593.60 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 31,021,458 *gallons*

Total nitrogen generated: 139,434.24 *lbs*

Total phosphorus generated: 16,627.02 *lbs*

Total potassium generated: 119,791.60 *lbs*

Total salt generated: 962,216.02 *lbs*

	31,385,508	*gallons applied*
+	6,222,000	*gallons exported*
-	6,586,050	*gallons imported*
=	31,021,458	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - D.W.N	Ground water
Domestic Well - D.W.S	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
08/19/2023	Process wastewater *Lagoon Water Import - Tiberio Azevedo*	6,586,050.00 *gal*	302.50	47.20	622.00		3,500

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	16,625.58	2,594.14	34,185.49	192,362.06
Total imports for all materials	16,625.58	2,594.14	34,185.49	192,362.06

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/27/2023	Corral solids	1,600.00 *ton*	Dry-weight	25.6		13,600.00	1,800.00	3,900.00		0.00
10/18/2023	Corral solids	1,500.00 *ton*	Dry-weight	72.8		19,900.00	4,600.00	10,800.00		0.00

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
08/16/2023	Process wastewater	6,222,000.00 *gal*	764.00	423.00	0.00	0.10	101.00	739.00		6,050

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	48,617.28	8,039.04	18,097.92	0.00
Process wastewater	39,674.05	5,244.18	38,370.79	314,131.67
Total exports for all materials	88,291.33	13,283.22	56,468.71	314,131.67

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	15	15	2	process wastewater	X045-X240-X043-XXXX
2	40	40	2	process wastewater	X045-X255-X003-XXXX
3	25	25	2	process wastewater	X045-X240-X041-XXXX
4	20	20	2	process wastewater	X045-X255-X002-XXXX
5	38	38	2	process wastewater	X045-X250-X029-XXXX
6	14	14	2	process wastewater	X045-X250-X028-XXXX
7	22	22	2	process wastewater	X045-X250-X028-XXXX
8	45	45	2	process wastewater	X045-X170-X003-XXXX
Lewis 30	30	30	0	none	X055-X011-X079-XXXX
Totals for areas that were used for application	219	219	16		
Totals for areas that were not used for application	30	30	0		
Land application area totals	249	249	16		

B. CROPS AND HARVESTS

1

Field name: 1

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	209.00 *ton*	Dry-weight		68.6	16,000.00	3,200.00	22,300.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.93	140.00	28.00	195.13	927.51

1

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	435.00 *ton*	Dry-weight		67.6	14,000.00	3,500.00	17,900.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.00	263.09	65.77	336.38	1,240.27

2

Field name: 2

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	563.00 *ton*	Dry-weight		72.5	18,000.00	3,400.00	23,700.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.08	139.34	26.32	183.47	688.97

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	1,123.00 *ton*	Dry-weight		67.2	14,600.00	3,700.00	18,300.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.08	268.89	68.14	337.03	1,141.87

3

Field name: 3

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	374.00 *ton*	Dry-weight		73.0	19,500.00	3,400.00	23,600.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.96	157.53	27.47	190.65	622.04

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	668.00 *ton*	Dry-weight		66.5	14,700.00	3,600.00	16,800.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.72	263.17	64.45	300.76	1,163.66

4

Field name: 4

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	296.00 *ton*	Dry-weight		73.1	19,500.00	3,500.00	23,400.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.80	155.27	27.87	186.32	652.92

4

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	534.00 *ton*	Dry-weight		67.1	14,300.00	3,800.00	17,300.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.70	251.23	66.76	303.94	1,124.39

5

Field name: 5

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	599.00 *ton*	Dry-weight		73.4	18,200.00	3,600.00	23,700.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.76	152.63	30.19	198.75	712.81

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	993.00 *ton*	Dry-weight		66.8	14,600.00	3,600.00	18,200.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.13	253.33	62.46	315.79	1,093.14

6

Field name: 6

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	213.00 *ton*	Dry-weight		73.5	19,300.00	3,700.00	23,000.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.21	155.63	29.84	185.46	620.90

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	367.00 *ton*	Dry-weight		66.6	14,700.00	3,700.00	17,800.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.21	257.41	64.79	311.70	1,103.20

7

Field name: 7

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	336.00 *ton*	Dry-weight		73.5	19,300.00	3,700.00	23,000.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.27	156.22	29.95	186.17	623.28

7

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	574.00 *ton*	Dry-weight		66.6	14,700.00	3,700.00	17,800.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.09	256.20	64.49	310.23	1,098.01

8

Field name: 8

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	687.00 *ton*	Dry-weight		72.8	17,600.00	3,600.00	21,200.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.27	146.17	29.90	176.07	689.32

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	1,199.00 *ton*	Dry-weight		68.7	15,100.00	3,900.00	18,600.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.64	251.86	65.05	310.24	1,117.52

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/16/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.13	5.72	50.46	433.27	320,495.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.72	1,885,800.00 *gal*
Application event totals		55.13	5.72	50.46	469.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	128.04	16.92	123.84	1,013.82	301,210.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.06	1,852,125.00 *gal*
Application event totals		128.04	16.92	123.84	1,049.88	

1 - 05/12/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.65	2,087,850.00 *gal*
Application event totals		0.00	0.00	0.00	40.65	

1 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.27	10.26	77.13	560.94	360,100.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.31	2,121,525.00 *gal*
Application event totals		90.27	10.26	77.13	602.25	

06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.44	1,717,425.00 *gal*
Application event totals		0.00	0.00	0.00	33.44	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.11	9.67	72.72	528.85	339,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.00	2,054,175.00 *gal*
Application event totals		85.11	9.67	72.72	568.85	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.00	6.13	54.00	463.70	343,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.44	1,717,425.00 *gal*
Application event totals		59.00	6.13	54.00	497.14	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.06	1,852,125.00 *gal*
Application event totals		0.00	0.00	0.00	36.06	

1 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.79	11.91	89.53	651.13	418,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.00	2,054,175.00 *gal*
Application event totals		104.79	11.91	89.53	691.13	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.34	2,020,500.00 *gal*
Application event totals		0.00	0.00	0.00	39.34	

2 - 11/16/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.87	10.67	80.20	583.27	998,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.52	5,960,475.00 *gal*
Application event totals		93.87	10.67	80.20	626.80	

04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.97	10.56	79.44	577.72	989,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.64	5,017,575.00 *gal*
Application event totals		92.97	10.56	79.44	614.36	

2 - 05/12/2023: Corn, silage

2 - 05/12/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.44	4,579,800.00 *gal*
Application event totals		0.00	0.00	0.00	33.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.56	9.61	72.25	525.44	899,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.69	4,613,475.00 *gal*
Application event totals		84.56	9.61	72.25	559.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.83	5,455,350.00 *gal*
Application event totals		0.00	0.00	0.00	39.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.46	9.60	72.17	524.86	898,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.69	4,613,475.00 *gal*
Application event totals		84.46	9.60	72.17	558.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.50	9.26	69.64	506.46	867,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.83	5,455,350.00 *gal*
Application event totals		81.50	9.26	69.64	546.29	

2 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.59	5,421,675.00 *gal*
Application event totals		0.00	0.00	0.00	39.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.89	10.44	78.51	571.01	977,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.85	5,320,650.00 *gal*
Application event totals		91.89	10.44	78.51	609.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.11	5,219,625.00 *gal*
Application event totals		0.00	0.00	0.00	38.11	

3 - 11/16/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.17	11.04	83.02	603.78	646,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.57	4,243,050.00 *gal*
Application event totals		97.17	11.04	83.02	653.35	

3 - 11/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.09	11.03	82.96	603.31	645,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.24	3,872,625.00 *gal*
Application event totals		97.09	11.03	82.96	648.55	

3 - 05/12/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.54	4,411,425.00 *gal*
Application event totals		0.00	0.00	0.00	51.54	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.66	6.19	54.60	468.84	578,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.74	3,401,175.00 *gal*
Application event totals		59.66	6.19	54.60	508.57	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.74	3,401,175.00 *gal*
Application event totals		0.00	0.00	0.00	39.74	

3 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.78	6.10	53.80	461.94	569,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.95	3,333,825.00 *gal*
Application event totals		58.78	6.10	53.80	500.89	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.66	6.19	54.60	468.84	578,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.34	3,367,500.00 *gal*
Application event totals		59.66	6.19	54.60	508.18	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.95	3,333,825.00 *gal*
Application event totals		0.00	0.00	0.00	38.95	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	149.59	19.77	144.68	1,184.43	586,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.34	3,367,500.00 *gal*
Application event totals		149.59	19.77	144.68	1,223.77	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.95	3,333,825.00 *gal*
Application event totals		0.00	0.00	0.00	38.95	

4 - 11/16/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

4 - 11/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.44	6.27	55.32	475.02	468,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.18	3,367,500.00 *gal*
Application event totals		60.44	6.27	55.32	524.20	

04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	146.72	19.39	141.90	1,161.71	460,200.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.18	3,367,500.00 *gal*
Application event totals		146.72	19.39	141.90	1,210.89	

4 - 05/12/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.47	6.59	58.10	498.85	492,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.80	2,862,375.00 *gal*
Application event totals		63.47	6.59	58.10	540.65	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.38	2,559,300.00 *gal*
Application event totals		0.00	0.00	0.00	37.38	

4 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.33	2,761,350.00 *gal*
Application event totals		0.00	0.00	0.00	40.33	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	155.43	20.54	150.32	1,230.63	487,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.87	2,592,975.00 *gal*
Application event totals		155.43	20.54	150.32	1,268.49	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.31	6.26	55.20	474.01	467,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.85	2,660,325.00 *gal*
Application event totals		60.31	6.26	55.20	512.86	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.87	2,592,975.00 *gal*
Application event totals		0.00	0.00	0.00	37.87	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.31	6.05	53.37	458.29	452,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.38	2,559,300.00 *gal*
Application event totals		58.31	6.05	53.37	495.67	

4 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.38	2,559,300.00 *gal*
Application event totals		0.00	0.00	0.00	37.38	

5 - 11/17/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.53	10.97	82.48	599.83	975,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.45	5,522,700.00 *gal*
Application event totals		96.53	10.97	82.48	642.28	

04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.54	10.86	81.63	593.68	965,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.67	5,421,675.00 *gal*
Application event totals		95.54	10.86	81.63	635.35	

5 - 05/13/2023: Corn, silage

Field name: 5

Crop: Corn, silage Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.16	10.13	76.18	554.02	901,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.08	5,084,925.00 *gal*
Application event totals		89.16	10.13	76.18	593.10	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.50	4,748,175.00 *gal*
Application event totals		0.00	0.00	0.00	36.50	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.20	4,579,800.00 *gal*
Application event totals		0.00	0.00	0.00	35.20	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.32	10.04	75.46	548.79	892,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.94	4,546,125.00 *gal*
Application event totals		88.32	10.04	75.46	583.74	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.20	4,579,800.00 *gal*
Application event totals		0.00	0.00	0.00	35.20	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.20	4,579,800.00 *gal*
Application event totals		0.00	0.00	0.00	35.20	

5 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	82.43	9.37	70.43	512.21	833,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.84	4,142,025.00 *gal*
Application event totals		82.43	9.37	70.43	544.04	

08/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	71.49	8.12	61.09	444.26	722,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.58	4,108,350.00 *gal*
Application event totals		71.49	8.12	61.09	475.84	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.84	4,142,025.00 *gal*
Application event totals		0.00	0.00	0.00	31.84	

6 - 11/17/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.62	8.82	66.32	482.34	289,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.45	1,986,825.00 *gal*
Application event totals		77.62	8.82	66.32	523.79	

6 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	123.88	16.38	119.82	980.90	272,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.05	1,919,475.00 *gal*
Application event totals		123.88	16.38	119.82	1,020.94	

6 - 05/13/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.64	7.00	52.67	383.04	229,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.37	2,222,550.00 *gal*
Application event totals		61.64	7.00	52.67	429.40	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.83	1,717,425.00 *gal*
Application event totals		0.00	0.00	0.00	35.83	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.92	7.26	54.62	397.22	238,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.83	1,717,425.00 *gal*
Application event totals		63.92	7.26	54.62	433.05	

6 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.42	1,650,075.00 *gal*
Application event totals		0.00	0.00	0.00	34.42	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	45.43	4.72	41.58	357.04	246,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.13	1,683,750.00 *gal*
Application event totals		45.43	4.72	41.58	392.17	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.64	7.00	52.67	383.04	229,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.13	1,683,750.00 *gal*
Application event totals		61.64	7.00	52.67	418.16	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.53	13.82	101.09	827.63	229,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.83	1,717,425.00 *gal*
Application event totals		104.53	13.82	101.09	863.46	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.64	1,852,125.00 *gal*
Application event totals		0.00	0.00	0.00	38.64	

7 - 11/17/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

7 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.14	8.99	67.62	491.75	463,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.50	3,502,200.00 *gal*
Application event totals		79.14	8.99	67.62	538.24	

04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	118.98	15.73	115.07	942.05	410,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.26	3,333,825.00 *gal*
Application event totals		118.98	15.73	115.07	986.31	

7 - 05/13/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	133.04	17.58	128.67	1,053.35	459,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.90	2,929,725.00 *gal*
Application event totals		133.04	17.58	128.67	1,092.24	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.42	2,592,975.00 *gal*
Application event totals		0.00	0.00	0.00	34.42	

7 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.87	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	34.87	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.91	9.08	68.27	496.53	467,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.32	2,660,325.00 *gal*
Application event totals		79.91	9.08	68.27	531.85	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.87	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	34.87	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.98	2,559,300.00 *gal*
Application event totals		0.00	0.00	0.00	33.98	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.09	8.42	63.31	460.42	433,500.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.53	2,525,625.00 *gal*
Application event totals		74.09	8.42	63.31	493.95	

7 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.84	5.38	47.45	407.41	442,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.08	2,491,950.00 *gal*
Application event totals		51.84	5.38	47.45	440.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.53	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	33.53	

8 - 11/17/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	101.55	11.54	86.77	631.05	1,215,325.00 *gal*
Lagoon Water	Process wastewater	31.31	4.89	64.39	362.30	558,200.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.24	6,970,725.00 *gal*
Application event totals		132.87	16.43	151.16	1,038.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	63.33	9.88	130.23	732.80	1,129,020.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.03	6,937,050.00 *gal*
Application event totals		63.33	9.88	130.23	777.82	

8 - 05/13/2023: Corn, silage

8 - 05/13/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.09	6,330,900.00 *gal*
Application event totals		0.00	0.00	0.00	41.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	48.09	7.50	98.88	556.41	857,255.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.87	6,297,225.00 *gal*
Application event totals		48.09	7.50	98.88	597.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	46.90	7.32	96.44	542.66	836,075.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.31	6,364,575.00 *gal*
Application event totals		46.90	7.32	96.44	583.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	46.95	7.33	96.54	543.24	836,975.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.09	6,330,900.00 *gal*
Application event totals		46.95	7.33	96.54	584.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.40	6,532,950.00 *gal*
Application event totals		0.00	0.00	0.00	42.40	

8 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	72.01	8.18	61.53	447.47	861,775.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.18	6,499,275.00 *gal*
Application event totals		72.01	8.18	61.53	489.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Import - Tiberio Azevedo	Process wastewater	71.00	8.07	60.66	441.16	849,625.00 *gal*
Lagoon Water	Process wastewater	56.43	7.46	54.58	446.82	398,253.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.87	6,297,225.00 *gal*
Application event totals		127.43	15.53	115.24	928.85	

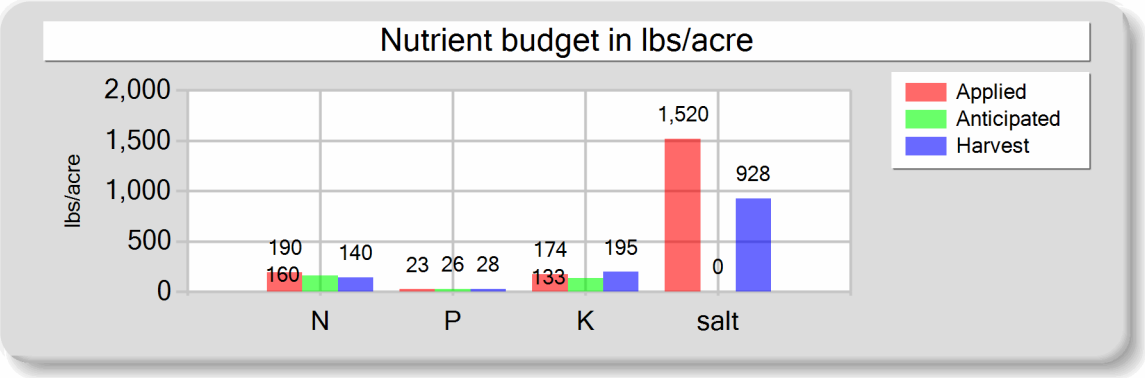
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.31	6,364,575.00 *gal*
Application event totals		0.00	0.00	0.00	41.31	

B. NUTRIENT BUDGET

1 - 11/16/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.17	22.65	174.30	1,447.09
Fresh water	0.00	0.00	0.00	72.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.17	22.65	174.30	1,519.88
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	140.00	28.00	195.13	927.51
Nutrient balance	50.17	-5.35	-20.83	592.36
Applied to removed ratio	1.36	0.81	0.89	1.64

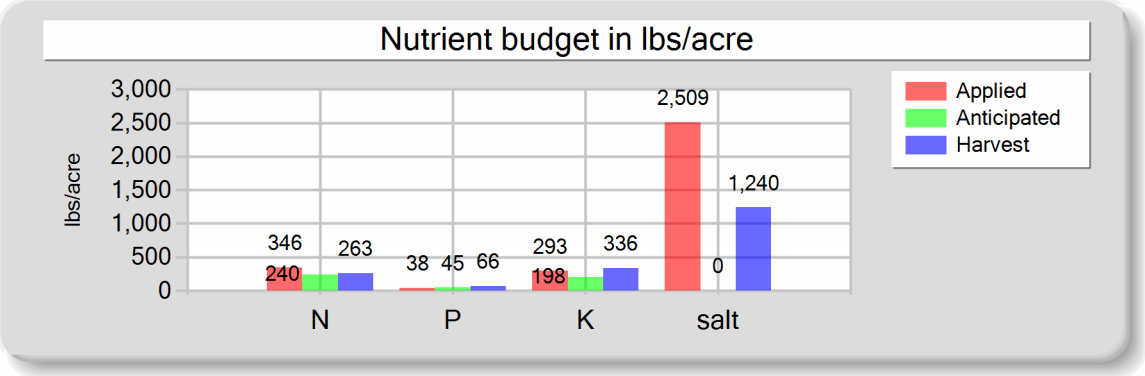
Fresh water applied	
3,737,925.00	*gallons*
137.66	*acre-inches*
9.18	*inches/acre*

Process wastewater applied	
621,705.00	*gallons*
22.90	*acre-inches*
1.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/12/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	339.17	37.96	293.38	2,204.62
Fresh water	0.00	0.00	0.00	304.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	346.17	37.96	293.38	2,508.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	263.09	65.77	336.38	1,240.27
Nutrient balance	83.08	-27.81	-43.00	1,268.60
Applied to removed ratio	1.32	0.58	0.87	2.02

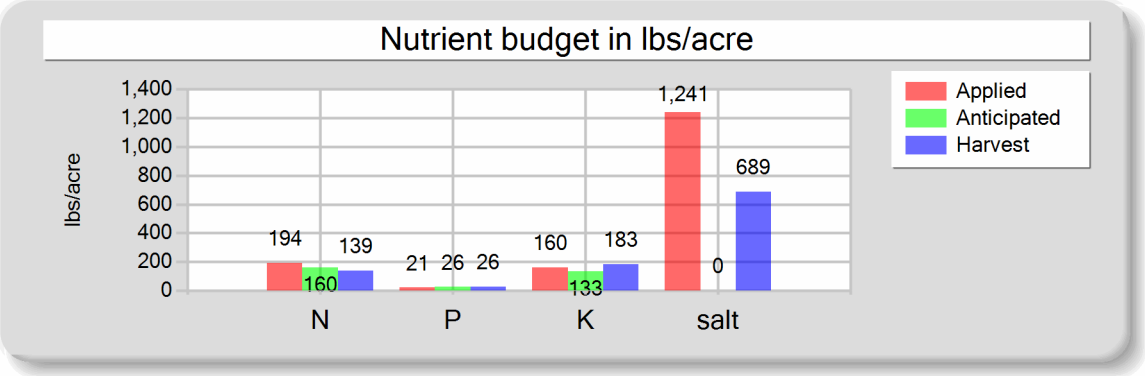
Fresh water applied	
15,625,200.00	*gallons*
575.42	*acre-inches*
38.36	*inches/acre*

Process wastewater applied	
1,460,600.00	*gallons*
53.79	*acre-inches*
3.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/16/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.84	21.23	159.64	1,161.00
Fresh water	0.00	0.00	0.00	80.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.84	21.23	159.64	1,241.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	139.34	26.32	183.47	688.97
Nutrient balance	54.50	-5.09	-23.83	552.19
Applied to removed ratio	1.39	0.81	0.87	1.80

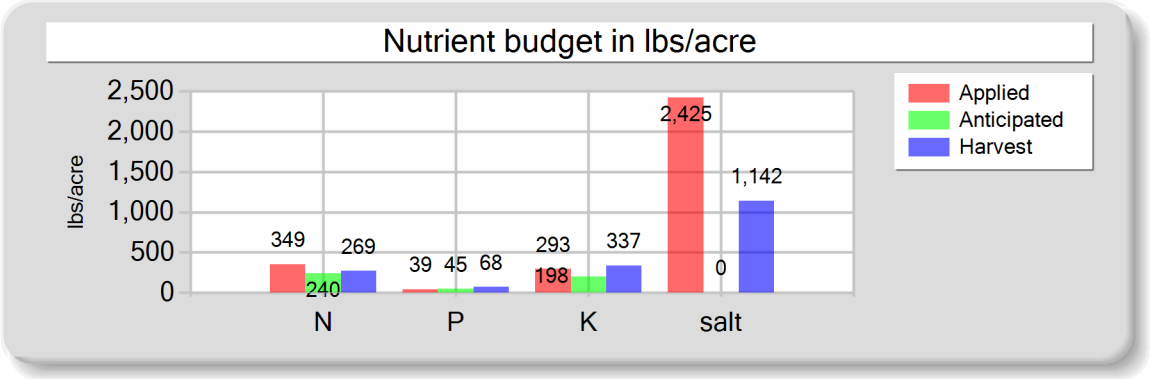
Fresh water applied	
10,978,050.00	*gallons*
404.28	*acre-inches*
10.11	*inches/acre*

Process wastewater applied	
1,987,500.00	*gallons*
73.19	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/12/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	342.42	38.91	292.57	2,127.77
Fresh water	0.00	0.00	0.00	297.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	349.42	38.91	292.57	2,424.80
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	268.89	68.14	337.03	1,141.87
Nutrient balance	80.53	-29.24	-44.47	1,282.94
Applied to removed ratio	1.30	0.57	0.87	2.12

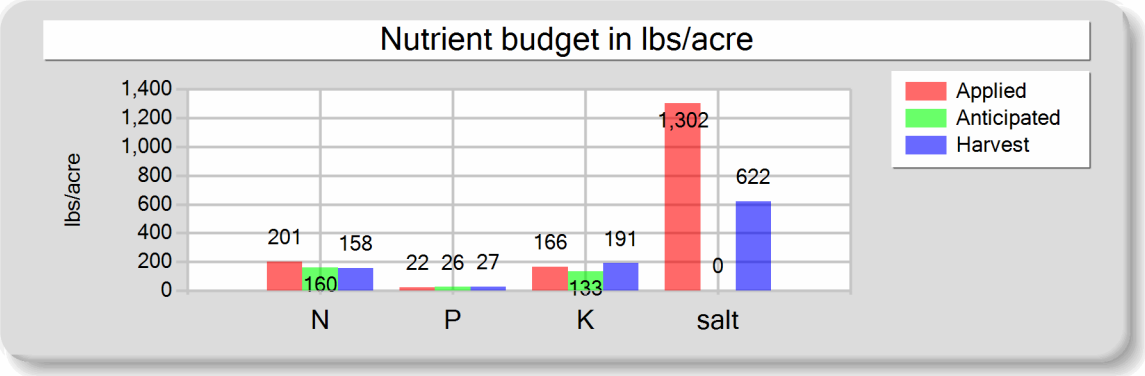
Fresh water applied	
40,679,400.00	*gallons*
1,498.08	*acre-inches*
37.45	*inches/acre*

Process wastewater applied	
3,642,500.00	*gallons*
134.14	*acre-inches*
3.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/16/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.25	22.07	165.97	1,207.09
Fresh water	0.00	0.00	0.00	94.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.25	22.07	165.97	1,301.90
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	157.53	27.47	190.65	622.04
Nutrient balance	43.73	-5.39	-24.68	679.87
Applied to removed ratio	1.28	0.80	0.87	2.09

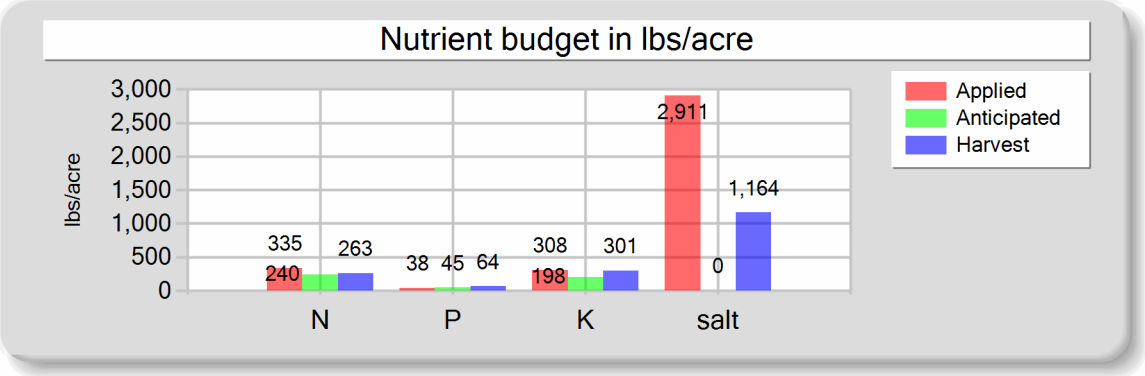
Fresh water applied	
8,115,675.00	*gallons*
298.87	*acre-inches*
11.95	*inches/acre*

Process wastewater applied	
1,291,500.00	*gallons*
47.56	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/12/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	327.68	38.26	307.68	2,584.04
Fresh water	0.00	0.00	0.00	326.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	334.68	38.26	307.68	2,910.59
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	263.17	64.45	300.76	1,163.66
Nutrient balance	71.52	-26.19	6.92	1,746.93
Applied to removed ratio	1.27	0.59	1.02	2.50

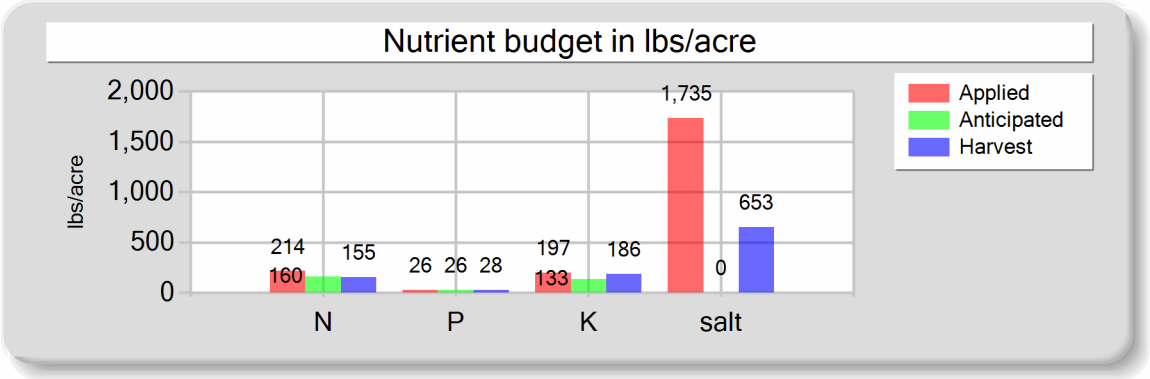
Fresh water applied	
27,950,250.00	*gallons*
1,029.31	*acre-inches*
41.17	*inches/acre*

Process wastewater applied	
2,312,000.00	*gallons*
85.14	*acre-inches*
3.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/16/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.16	25.67	197.22	1,636.73
Fresh water	0.00	0.00	0.00	98.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.16	25.67	197.22	1,735.09
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	155.27	27.87	186.32	652.92
Nutrient balance	58.90	-2.20	10.90	1,082.17
Applied to removed ratio	1.38	0.92	1.06	2.66

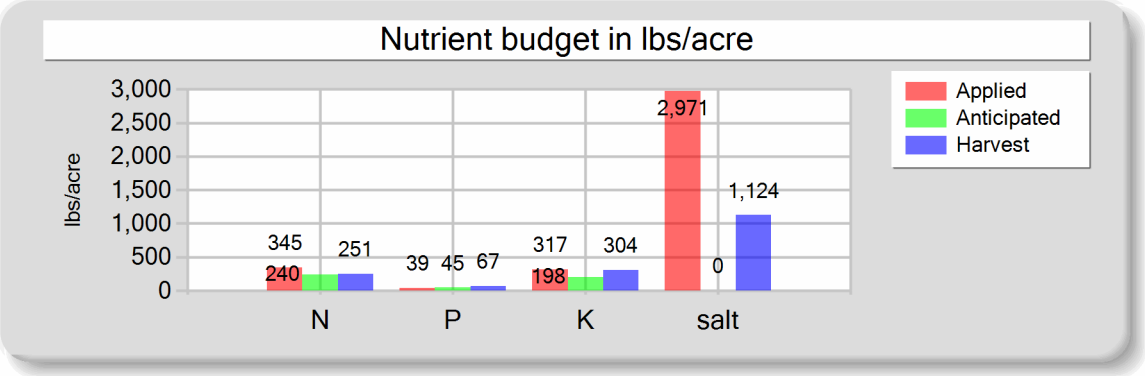
Fresh water applied	
6,735,000.00	*gallons*
248.03	*acre-inches*
12.40	*inches/acre*

Process wastewater applied	
928,700.00	*gallons*
34.20	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 05/12/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	337.53	39.45	316.99	2,661.77
Fresh water	0.00	0.00	0.00	308.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.53	39.45	316.99	2,970.61
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	251.23	66.76	303.94	1,124.39
Nutrient balance	93.30	-27.31	13.06	1,846.22
Applied to removed ratio	1.37	0.59	1.04	2.64

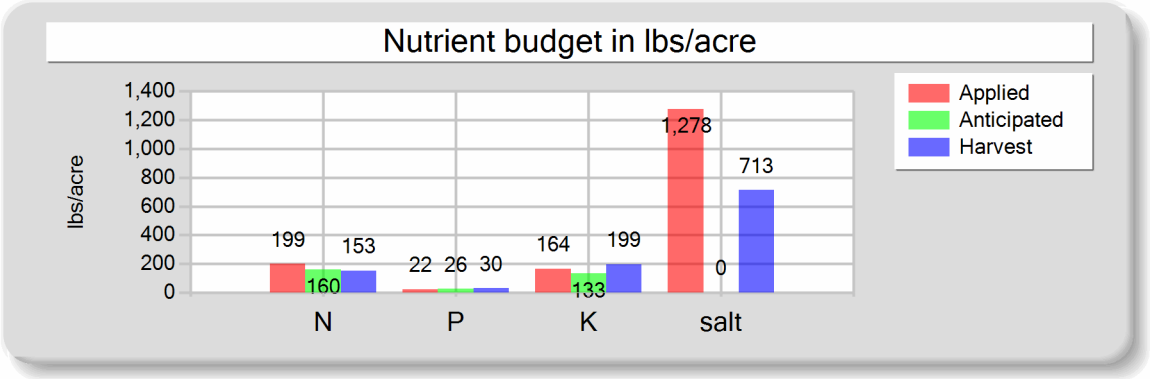
Fresh water applied	
21,147,900.00	*gallons*
778.81	*acre-inches*
38.94	*inches/acre*

Process wastewater applied	
1,899,000.00	*gallons*
69.93	*acre-inches*
3.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/17/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.07	21.82	164.11	1,193.51
Fresh water	0.00	0.00	0.00	84.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.07	21.82	164.11	1,277.63
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.63	30.19	198.75	712.81
Nutrient balance	46.44	-8.37	-34.64	564.82
Applied to removed ratio	1.30	0.72	0.83	1.79

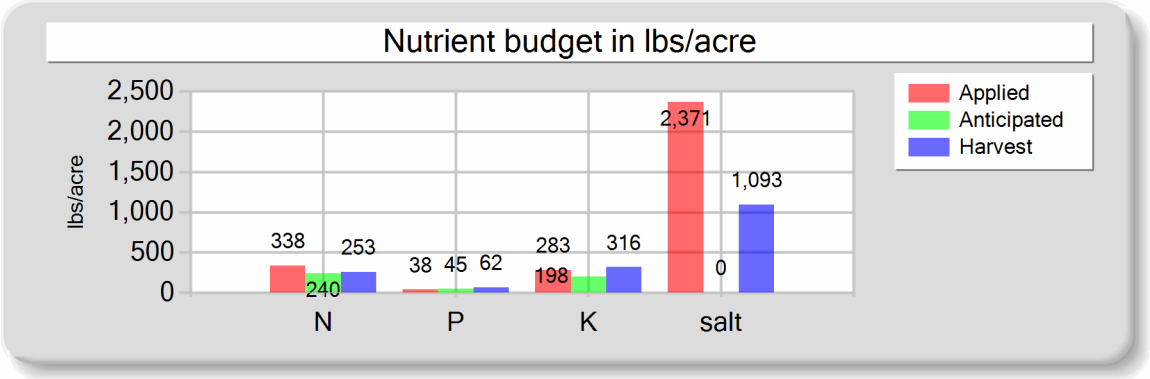
Fresh water applied	
10,944,375.00	*gallons*
403.04	*acre-inches*
10.61	*inches/acre*

Process wastewater applied	
1,941,000.00	*gallons*
71.48	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 05/13/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	331.40	37.66	283.15	2,059.28
Fresh water	0.00	0.00	0.00	311.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	338.40	37.66	283.15	2,370.66
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	253.33	62.46	315.79	1,093.14
Nutrient balance	85.07	-24.81	-32.64	1,277.52
Applied to removed ratio	1.34	0.60	0.90	2.17

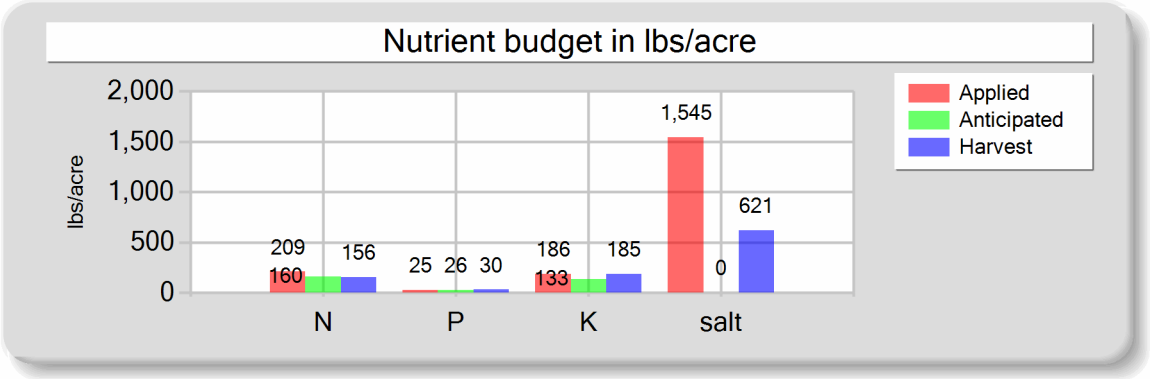
Fresh water applied	
40,511,025.00	*gallons*
1,491.88	*acre-inches*
39.26	*inches/acre*

Process wastewater applied	
3,349,000.00	*gallons*
123.33	*acre-inches*
3.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/17/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.51	25.20	186.14	1,463.24
Fresh water	0.00	0.00	0.00	81.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.51	25.20	186.14	1,544.73
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	155.63	29.84	185.46	620.90
Nutrient balance	52.88	-4.64	0.67	923.84
Applied to removed ratio	1.34	0.84	1.00	2.49

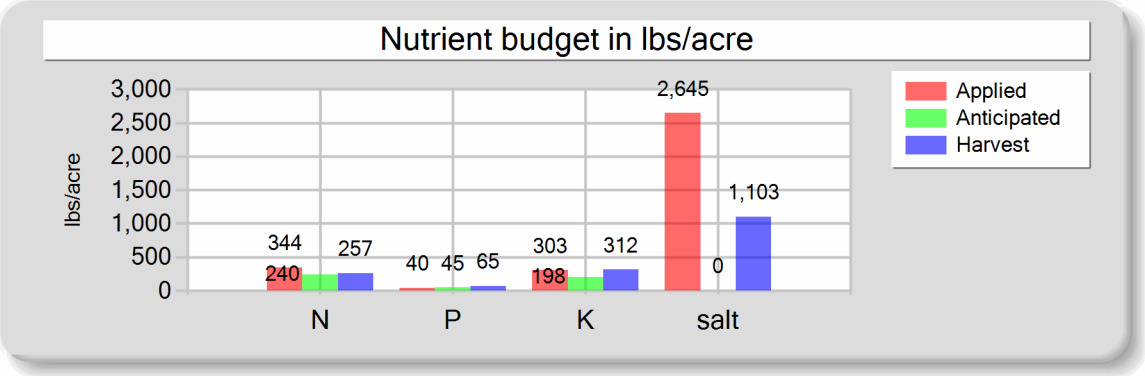
Fresh water applied	
3,906,300.00	*gallons*
143.86	*acre-inches*
10.28	*inches/acre*

Process wastewater applied	
561,000.00	*gallons*
20.66	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 05/13/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	337.17	39.80	302.63	2,347.97
Fresh water	0.00	0.00	0.00	297.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.17	39.80	302.63	2,645.14
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	257.41	64.79	311.70	1,103.20
Nutrient balance	86.75	-24.99	-9.07	1,541.94
Applied to removed ratio	1.34	0.61	0.97	2.40

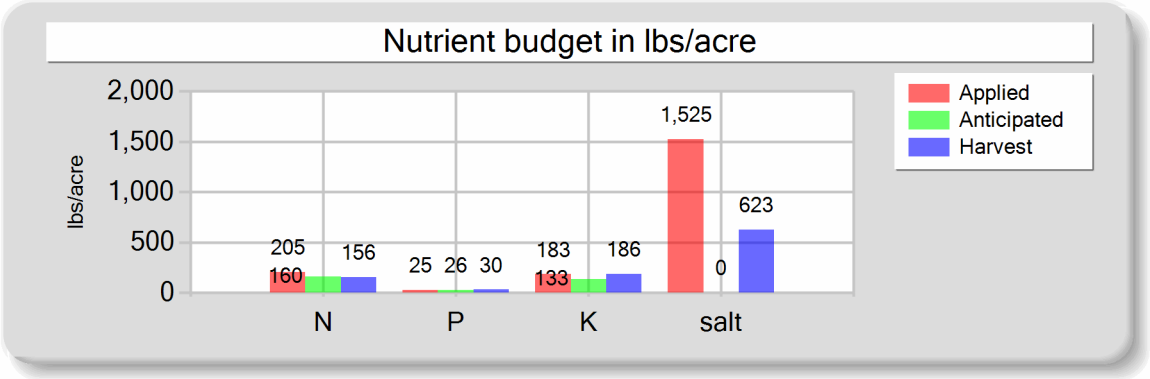
Fresh water applied	
14,244,525.00	*gallons*
524.58	*acre-inches*
37.47	*inches/acre*

Process wastewater applied	
1,173,000.00	*gallons*
43.20	*acre-inches*
3.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/17/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	198.11	24.72	182.69	1,433.79
Fresh water	0.00	0.00	0.00	90.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.11	24.72	182.69	1,524.55
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	156.22	29.95	186.17	623.28
Nutrient balance	48.89	-5.23	-3.49	901.27
Applied to removed ratio	1.31	0.83	0.98	2.45

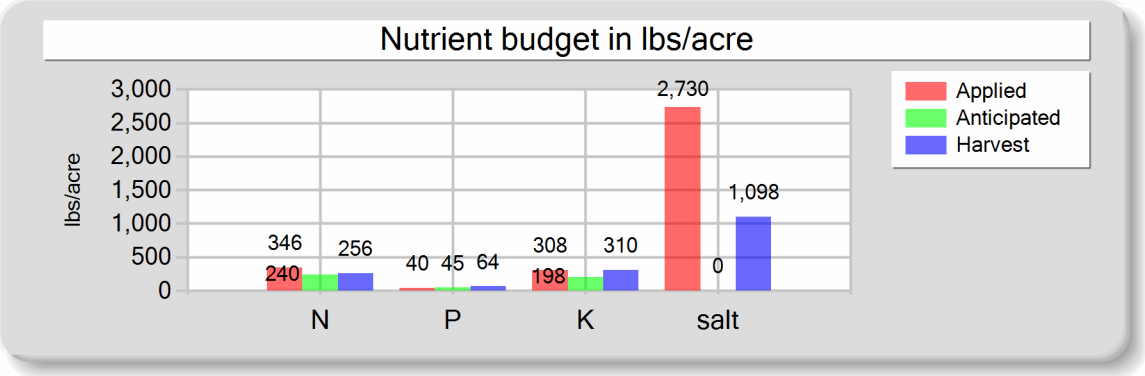
Fresh water applied	
6,836,025.00	*gallons*
251.75	*acre-inches*
11.44	*inches/acre*

Process wastewater applied	
873,500.00	*gallons*
32.17	*acre-inches*
1.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 05/13/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	338.87	40.47	307.69	2,417.70
Fresh water	0.00	0.00	0.00	312.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	345.87	40.47	307.69	2,730.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	256.20	64.49	310.23	1,098.01
Nutrient balance	89.67	-24.02	-2.54	1,632.20
Applied to removed ratio	1.35	0.63	0.99	2.49

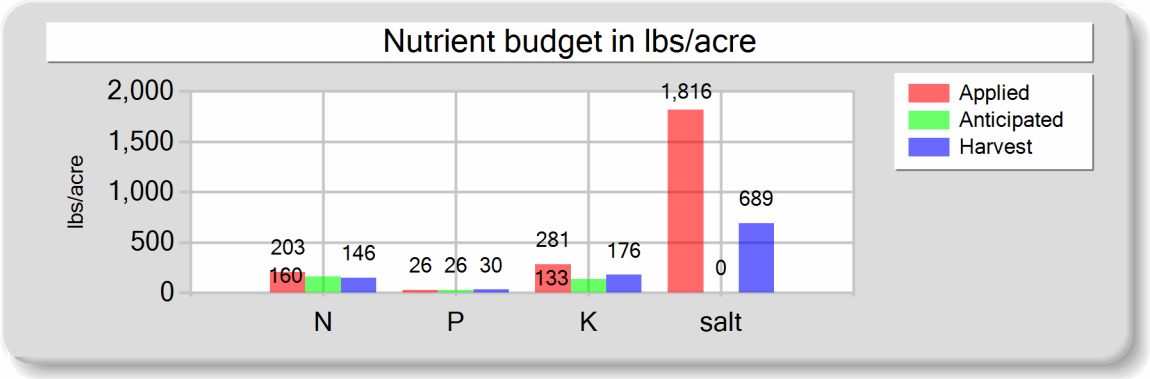
Fresh water applied	
23,538,825.00	*gallons*
866.85	*acre-inches*
39.40	*inches/acre*

Process wastewater applied	
1,802,000.00	*gallons*
66.36	*acre-inches*
3.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/17/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.20	26.31	281.38	1,726.15
Fresh water	0.00	0.00	0.00	90.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.20	26.31	281.38	1,816.42
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	146.17	29.90	176.07	689.32
Nutrient balance	57.03	-3.59	105.32	1,127.10
Applied to removed ratio	1.39	0.88	1.60	2.64

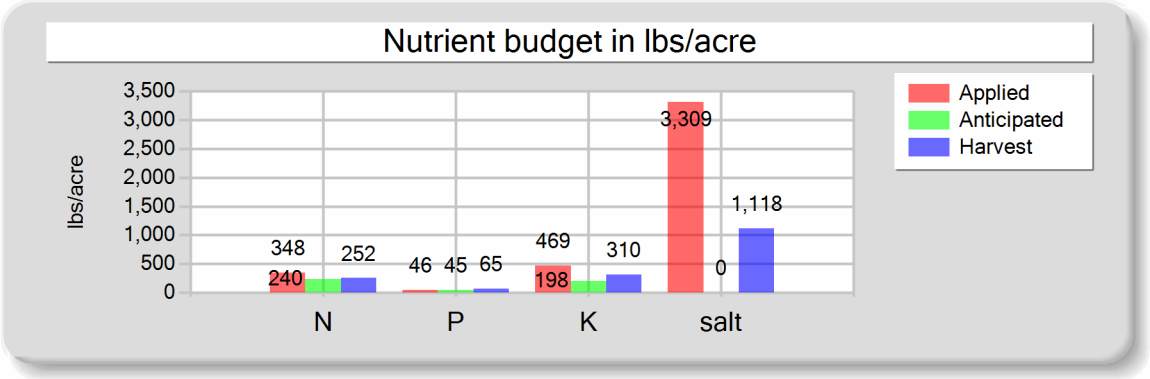
Fresh water applied	
13,907,775.00	*gallons*
512.18	*acre-inches*
11.38	*inches/acre*

Process wastewater applied	
2,902,545.00	*gallons*
106.89	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 05/13/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	341.38	45.86	468.63	2,977.76
Fresh water	0.00	0.00	0.00	331.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	348.38	45.86	468.63	3,308.89
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	251.86	65.05	310.24	1,117.52
Nutrient balance	96.52	-19.19	158.39	2,191.37
Applied to removed ratio	1.38	0.70	1.51	2.96

Fresh water applied	
51,017,625.00	*gallons*
1,878.81	*acre-inches*
41.75	*inches/acre*

Process wastewater applied	
4,639,958.00	*gallons*
170.87	*acre-inches*
3.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 25.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	1,800.00	3,900.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,900.00	4,600.00	10,800.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	784.00	402.00	0.00	1.90	103.00	625.00								8,240.00	4,240
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	450.00	297.00	0.00	0.60	51.20	385.00								5,940.00	2,800
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	304.00	190.00	0.00	5.20	32.10	283.00								4,080.00	2,430
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water Import - Tiberio Azevedo

Sample and source description: Lagoon Water Import - Tiberio Azevedo

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	302.00	203.00	0.00	0.50	47.20	622.00								4,790.00	3,500
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	764.00	423.00	0.00	0.10	101.00	739.00								8,870.00	6,050
DL	1.00	0.50	1.00	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - D.W.N

D.W.N

Sample description: D.W.N

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.90		3.90								1,040.00	
DL	1.00		0.10								10.00	

Domestic Well - D.W.S

D.W.S

Sample description: D.W.S

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.90		3.90								1,010.00	
DL	1.00		0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										31.40	35
DL	1.00										10.00	10

D. SOIL ANALYSES

1

B6 N

Sample and source description: B6 N

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,234.00				
DL			2.00				

2

B6 N

Sample and source description: B6 N

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,234.00				
DL			2.00				

3

B6 N

Sample and source description: B6 N

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,234.00				
DL			2.00				

4

4

B6 N

Sample and source description: B6 N

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,234.00				
DL			2.00				

5

B6 S

Sample and source description: B6 S

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,572.00				
DL			2.00				

6

B6 S

Sample and source description: B6 S

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,572.00				
DL			2.00				

7

7

B6 S

Sample and source description: B6 S

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,572.00				
DL			2.00				

8

B6 AUG

Sample and source description: B6 AUG

Sample date: 05/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			468.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

1 - 11/16/2022: Oats, silage-soft dough

1

Sample and source description: 1

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,000.00	3,200.00	22,300.00		10.60
DL	100.00	100.00	100.00		0.01

1 - 05/12/2023: Corn, silage

1 - 05/12/2023: Corn, silage

1

Sample and source description: 1

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	3,500.00	17,900.00		6.60
DL	100.00	100.00	100.00		0.01

2 - 11/16/2022: Oats, silage-soft dough

2

Sample and source description: 2

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,000.00	3,400.00	23,700.00		8.90
DL	100.00	100.00	100.00		0.01

2 - 05/12/2023: Corn, silage

2

Sample and source description: 2

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	3,700.00	18,300.00		6.20
DL	100.00	100.00	100.00		0.01

3 - 11/16/2022: Oats, silage-soft dough

3

Sample and source description: 3

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	3,400.00	23,600.00		7.70
DL	100.00	100.00	100.00		0.01

3 - 05/12/2023: Corn, silage

3

Sample and source description: 3

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	3,600.00	16,800.00		6.50
DL	100.00	100.00	100.00		0.01

4 - 11/16/2022: Oats, silage-soft dough

4

Sample and source description: 4

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	3,500.00	23,400.00		8.20
DL	100.00	100.00	100.00		0.01

4 - 05/12/2023: Corn, silage

4

Sample and source description: 4

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,800.00	17,300.00		6.40
DL	100.00	100.00	100.00		0.01

5 - 11/17/2022: Oats, silage-soft dough

5

Sample and source description: 5

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	3,600.00	23,700.00		8.50
DL	100.00	100.00	100.00		0.01

5 - 05/13/2023: Corn, silage

5

Sample and source description: 5

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	3,600.00	18,200.00		6.30
DL	100.00	100.00	100.00		0.01

6 - 11/17/2022: Oats, silage-soft dough

6 & 7

Sample and source description: 6 & 7

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	3,700.00	23,000.00		7.70
DL	100.00	100.00	100.00		0.01

6 - 05/13/2023: Corn, silage

6 & 7

Sample and source description: 6 & 7

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	3,700.00	17,800.00		6.30
DL	100.00	100.00	100.00		0.01

7 - 11/17/2022: Oats, silage-soft dough

6 & 7

Sample and source description: 6 & 7

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	3,700.00	23,000.00		7.70
DL	100.00	100.00	100.00		0.01

7 - 05/13/2023: Corn, silage

6 & 7

Sample and source description: 6 & 7

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	3,700.00	17,800.00		6.30
DL	100.00	100.00	100.00		0.01

8 - 11/17/2022: Oats, silage-soft dough

8A

Sample and source description: 8A

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	3,500.00	22,500.00		9.30
DL	100.00	100.00	100.00		0.01

8B

Sample and source description: 8B

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	3,600.00	21,200.00		8.30
DL	100.00	100.00	100.00		0.01

8 - 05/13/2023: Corn, silage

8 - 05/13/2023: Corn, silage

8A

Sample and source description: 8A

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	3,900.00	18,600.00		6.70
DL	100.00	100.00	100.00		0.01

8b

Sample and source description: 8b

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	3,700.00	17,300.00		6.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

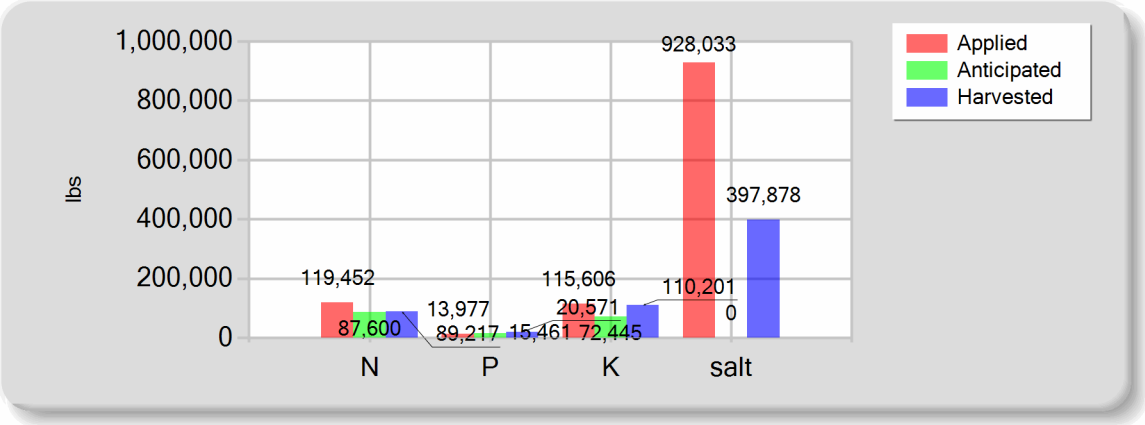
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

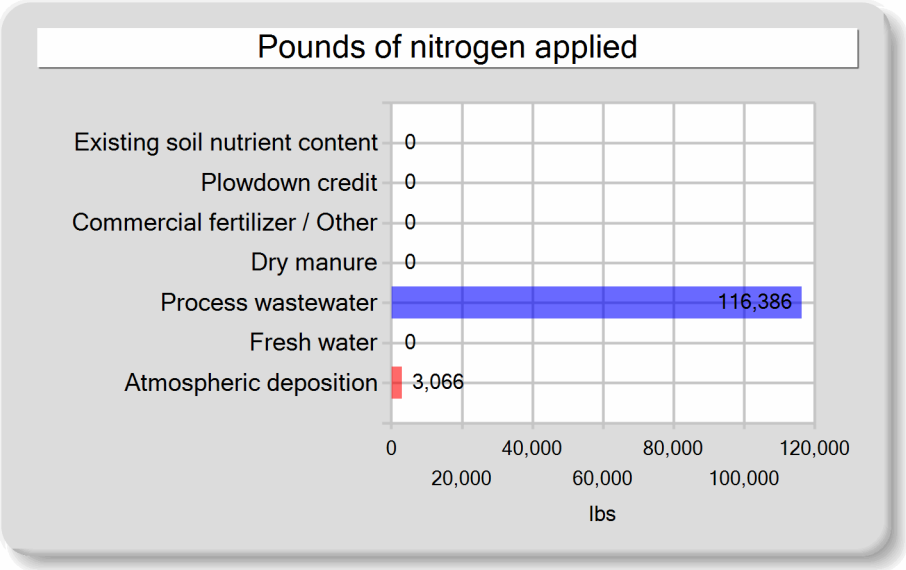
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

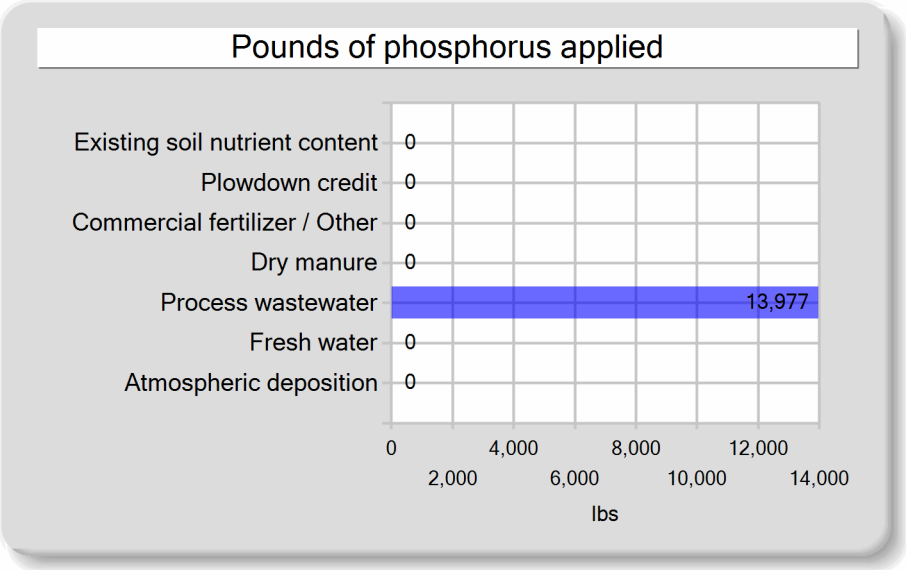
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	116,385.77	13,976.97	115,606.29	840,446.41
Fresh water	0.00	0.00	0.00	87,586.25
Atmospheric deposition	3,066.00	0.00	0.00	0.00
Total nutrients applied	119,451.77	13,976.97	115,606.29	928,032.66
Anticipated crop nutrient removal	87,600.00	15,461.40	72,445.20	0.00
Actual crop nutrient removal	89,216.55	20,571.50	110,200.59	397,878.33
Nutrient balance	30,235.22	-6,594.53	5,405.70	530,154.32
Applied to removed ratio	1.34	0.68	1.05	2.33

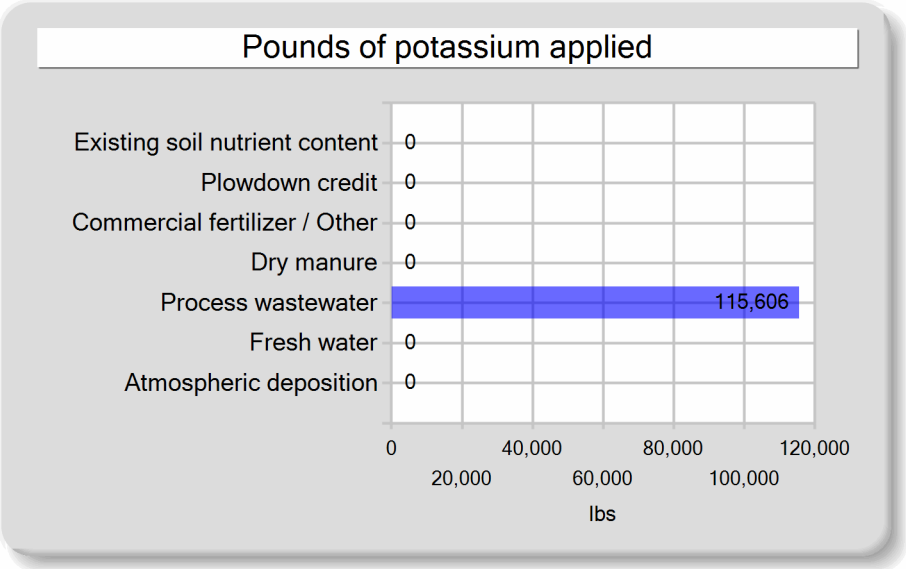
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

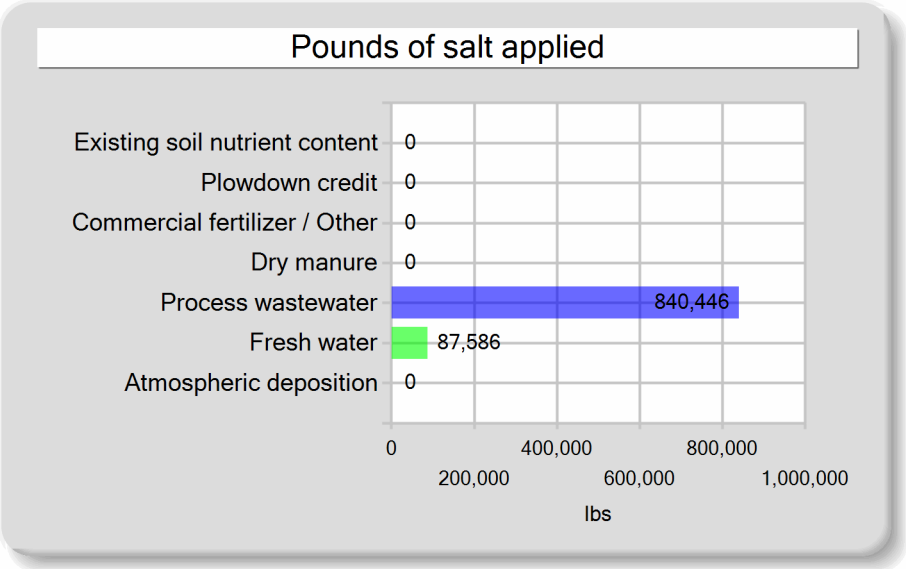


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

ALL SOIL PHOSPHORUS Results reported are expressed as pounds per acre of P2O5 or OSLEN*

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Michael Borba
Michael Borba (Jun 19, 2024 12:07 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Mike Borba	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/19/2024	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Mike or Emanuel Borba**

Name of Dairy Facility: **B-6 Dairy**

Facility Address: **22014 Turner Road, Hilmar, CA 95324**

Contact Person Name and Phone Number: **Mike or Emanuel Borba (209) 634-4930**

Manure Hauler Information:

Name of Hauling Company/Person: **Five Star Custom Spreading**

Address of Hauling Company /Person: **219 Jensen Road, Gustine, CA 95322**

Contact Person and Phone Number: **Eric Rocha (209) 678-2917**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
B & B Farms	P.O. Box 922	Hilmar	95324	(209) 634-4930

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
Merced County			0045-0330-0020-0000; 0045-0330-0021-0000

Dates Hauled: **5/27/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,600** Tons

Manure Solids Content (if amount reported in tons): **74.4%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Michael Borba (Jun 19, 2024 12:07 PDT) Date: 06/19/2024

Hauler's Signature: _____ Date: 5/27/2023

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator:	**Mike or Emanuel Borba**
Name of Dairy Facility:	**B-6 Dairy**
Facility Address:	**22014 Turner Road, Hilmar, CA 95324**
Contact Person Name and Phone Number:	**Mike or Emanuel Borba (209) 634-4930**

Manure Hauler Information:

Name of Hauling Company/Person:	**Five Star Custom Spreading**
Address of Hauling Company /Person:	**219 Jensen Road, Gustine, CA 95322**
Contact Person and Phone Number:	**Eric Rocha (209) 678-2917**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
B & B Farms	**P.O. Box 922**	**Hilmar**	**95324**	**(209) 634-4930**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
Merced County			**0045-0330-0020-0000; 0045-0330-0021-0000**

Dates Hauled: **10/18/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,500** Tons

Manure Solids Content (if amount reported in tons): **27.2%**

Method used to determine amount of manure: **Number of Truckloads x Average Truck Weight**

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Michael Borba (Jun 19, 2024 12:07 PDT) Date: 06/19/2024

Hauler's Signature: Eric Roth Date: 10/18/2023

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Mike or Emanuel Borba**

Name of Dairy Facility: **B-6 Dairy**

Facility Address: **22014 Turner Road, Hilmar, CA 95324**

Contact Person Name and Phone Number: **Mike or Emanuel Borba (209) 634-4930**

Manure Hauler Information:

Name of Hauling Company/Person: **B-6 Dairy**

Address of Hauling Company /Person: **22014 Turner Road, Hilmar, CA 95324**

Contact Person and Phone Number: **Mike or Emanuel Borba (209) 634-4930**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Bob Livak	**22339 Crane Avenue**	**Hilmar**	**95324**	**(209) 668-8319**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
			0045-0240-0020-000

Dates Hauled: **5/29/23, 6/18/23, 7/6/23, 7/21/23 & 8/16/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 6,222,000 Gallons

Method used to determine volume of process wastewater: _____

Lagoon Pump GPM x Total Run Time

Written Agreement:

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Michael Borba
Michael Borba (Jun 19, 2024 12:07 PDT) Date: 06/19/2024

Hauler's Signature: Michael Borba
Michael Borba (Jun 19, 2024 12:07 PDT) Date: 06/19/2024

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: B-6 Dairy

Received: 11/16/2023 15:45
Reported: 11/22/2023 09:46

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0907-01	D.W.N	Ag Water	T. Madruga		11/15/2023 15:00
23K0907-02	D.W.S	Ag Water	T. Madruga		11/15/2023 16:00

Default Cooler Temperature on Receipt °C: 0.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: B-6 Dairy

Received: 11/16/2023 15:45
Reported: 11/22/2023 09:46

Sample Results

Sample: D.W.N
23K0907-01 (Water)

Sampled: 11/15/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.04**	mmhos/cm	0.01	1		11/20/23 15:17	SM 2510 B		BEK0618
Electrical Conductivity umhos	**1040**	umhos/cm	10.0	1		11/20/23 15:17	SM 2510 B		BEK0618
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:00	Field		BEK0569
Nitrate Nitrogen as NO3N	**3.9**	mg/L	0.1	1	10	11/17/23 03:19	EPA 300.0		BEK0489
Temperature	**25.0**	units	0.0	1		11/20/23 15:17	SM 4500-H+	H	BEK0618
pH	**8.3**	units	1.0	1		11/20/23 15:17	SM 4500-H+	H	BEK0618

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Ranch: B-6 Dairy

Received: 11/16/2023 15:45
Reported: 11/22/2023 09:46

Sample Results
(Continued)

Sample: D.W.S
23K0907-02 (Water)

Sampled: 11/15/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.01**	mmhos/cm	0.01	1		11/20/23 15:24	SM 2510 B		BEK0618
Electrical Conductivity umhos	**1010**	umhos/cm	10.0	1		11/20/23 15:24	SM 2510 B		BEK0618
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 16:00	Field		BEK0569
Nitrate Nitrogen as NO3N	**3.9**	mg/L	0.1	1	10	11/17/23 03:38	EPA 300.0		BEK0489
Temperature	**25.0**	units	0.0	1		11/20/23 15:24	SM 4500-H+	H	BEK0618
pH	**8.2**	units	1.0	1		11/20/23 15:24	SM 4500-H+	H	BEK0618

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R & J Avila Inc
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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: B-6 Dairy

Received: 11/16/2023 15:45
Reported: 11/22/2023 09:46

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0489										
Blank (BEK0489-BLK1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0489-BLK2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0489-BLK3)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0489-BLK4)					Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0489-BS1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEK0489-BS2)					Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000		107	90-110		
LCS (BEK0489-BS3)					Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		105	90-110		
Duplicate (BEK0489-DUP1)		**Source: 23K0831-01**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	1.5		0.1	mg/L		1.5			0.539	10
Duplicate (BEK0489-DUP2)		**Source: 23K0908-02**			Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	0.8		0.1	mg/L		0.8			0.397	10
Duplicate (BEK0489-DUP3)		**Source: 23K0941-02**			Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	1.8		0.1	mg/L		1.8			0.392	10
Matrix Spike (BEK0489-MS1)		**Source: 23K0831-01**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.5	106	90-110		
Matrix Spike (BEK0489-MS2)		**Source: 23K0908-02**			Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	6.0		0.1	mg/L	5.000	0.8	105	90-110		
Matrix Spike (BEK0489-MS3)		**Source: 23K0941-02**			Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	7.1		0.1	mg/L	5.000	1.8	106	90-110		
Reference (BEK0489-SRM1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0489-SRM2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: B-6 Dairy

Received: 11/16/2023 15:45
Reported: 11/22/2023 09:46

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0489 (Continued)										
Reference (BEK0489-SRM3)					Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0489-SRM4)					Prepared & Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Account# 00-0019840
Account Manager: Ben Nydam
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Ranch: B-6 Dairy

Received: 11/16/2023 15:45
Reported: 11/22/2023 09:46

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0618										
Blank (BEK0618-BLK1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0618-BLK2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.1		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0618-BLK3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Duplicate (BEK0618-DUP1)		**Source: 23K0907-01**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1.03		0.01	mmhos/cm		1.04			0.907	10
Electrical Conductivity umhos	1030		10.0	umhos/cm		1040			0.907	10
pH	8.3		1.0	units		8.3			0.242	10
Duplicate (BEK0618-DUP2)		**Source: 23K0930-04**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	0.57		0.01	mmhos/cm		0.57			0.264	10
Electrical Conductivity umhos	567		10.0	umhos/cm		569			0.264	10
pH	8.1		1.0	units		8.1			0.248	10
Reference (BEK0618-SRM1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	554			umhos/cm	538.0		103	90-110		
Reference (BEK0618-SRM2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEK0618-SRM3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0618-SRM4)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0618-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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LABORATORY INC

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Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: B-6 Dairy

Received: 11/16/2023 15:45
Reported: 11/22/2023 09:46

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0618 (Continued)										
Reference (BEK0618-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEK0618-SRM6)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0618-SRM7)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0618-SRM8)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		

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11/16/23 15:45 23K0907

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 2 No. of Bottles: 2

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT B-6 Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	D.W.N	11/15/2023	1500	0.4	ND
2	D.W.S	11/15/2023	1600	-0.3	ND
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/15/2023 @ 1600	11/16/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/16/2023 @9am	11/16/23 3:25
Third		DCF	11/16/23 15:45	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:45 23K0907

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEMI-OUT

11/16/23 15:45 23K0907

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1								
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

DRAFT
SEMI-FINAL